

Class/ Index Number	Centre Number/ 'O' Level Index Number	Name
/	/	SOLUTIONS



新加坡海星中学
MARIS STELLA HIGH SCHOOL
PRELIMINARY EXAMINATION
SECONDARY FOUR

ADDITIONAL MATHEMATICS

4049/2

Paper 2

25 August 2026

Candidates answer on the Question Paper.

2 hours 15 minutes

READ THESE INSTRUCTIONS FIRST

Write your class, index number and name in the spaces at the top of this page.

Write in dark blue or black pen.

You may use an HB pencil for any diagrams or graphs.

Do not use staples, paper clips, glue or correction fluid.

Answer **all** the questions.

Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.

The use of an approved scientific calculator is expected, where appropriate.

You are reminded of the need for clear presentation in your answers.

The number of marks is given in brackets [] at the end of each question or part question.

The total of the marks for this paper is 90.

For Examiners' Use

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8
/4	/5	/6	/7	/4	/9	/10	/9
Q9	Q10	Q11	Q12	SUBTOTAL		90	
/8	/9	/9	/10	Statement			
				Presentation			
				Units			
				Rounding Off			

Mathematical Formulae

1. ALGEBRA

Quadratic Equation

For the equation $ax^2 + bx + c = 0$,

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Binomial expansion

$$(a + b)^n = a^n + \binom{n}{1} a^{n-1} b + \binom{n}{2} a^{n-2} b^2 + \dots + \binom{n}{r} a^{n-r} b^r + \dots + b^n,$$

where n is a positive integer and $\binom{n}{r} = \frac{n!}{r!(n-r)!} = \frac{n(n-1)\dots(n-r+1)}{r!}$

2. TRIGONOMETRY

Identities

$$\sin^2 A + \cos^2 A = 1$$

$$\sec^2 A = 1 + \tan^2 A$$

$$\operatorname{cosec}^2 A = 1 + \cot^2 A$$

$$\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$$

$$\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$$

$$\tan(A \pm B) = \frac{\tan A \pm \tan B}{1 \mp \tan A \tan B}$$

$$\sin 2A = 2 \sin A \cos A$$

$$\cos 2A = \cos^2 A - \sin^2 A = 2 \cos^2 A - 1 = 1 - 2 \sin^2 A$$

$$\tan 2A = \frac{2 \tan A}{1 - \tan^2 A}$$

Formulae for ΔABC

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$a^2 = b^2 + c^2 - 2bc \cos A$$

$$\Delta = \frac{1}{2} bc \sin A$$

1 Do not use a calculator in this question.

In triangle ABC , $AB = (6 + \sqrt{5})$ cm and $\angle ABC = 30^\circ$. Given that the triangle has an area of $\frac{1}{4}(32 - 5\sqrt{5})$ cm², find the exact length of BC . Give your answer in the form $(a + b\sqrt{5})$, where a and b are integers. [4]

$\begin{aligned} \frac{1}{2}(6 + \sqrt{5})(BC) \sin 30^\circ &= \frac{1}{4}(32 - 5\sqrt{5}) \\ \frac{1}{2}(6 + \sqrt{5})(BC) \left(\frac{1}{2}\right) &= \frac{1}{4}(32 - 5\sqrt{5}) \\ BC &= \frac{32 - 5\sqrt{5}}{6 + \sqrt{5}} \times \frac{6 - \sqrt{5}}{6 - \sqrt{5}} \\ &= \frac{192 - 32\sqrt{5} - 30\sqrt{5} + 25}{31} \\ &= \frac{217 - 62\sqrt{5}}{31} \\ &= (7 - 2\sqrt{5}) \text{ cm} \end{aligned}$	
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- 2 In a sports meet, the height, h metres, of a javelin thrown by an athlete is modelled by the formula,
 $h = -\frac{3}{100}(w^2 - 36w - 76)$, where w is the horizontal distance travelled by the javelin in metres.

- (a) By expressing h in the form $A(w-B)^2 + C$, where A , B and C are constants, find the maximum height reached by the javelin and state the corresponding horizontal distance when this occurs. [3]

$$\begin{aligned} h &= -\frac{3}{100}(w^2 - 36w - 76) \\ &= -\frac{3}{100}[(w-18)^2 - 18^2 - 76] \\ &= -\frac{3}{100}(w-18)^2 + 12 \end{aligned}$$

Maximum height of javelin = 12 m when the horizontal distance is at 18 m.

- (b) Find the horizontal distance covered by the javelin before it hits the ground. [2]

$$\begin{aligned} \text{When } h &= 0, \\ -\frac{3}{100}(w-18)^2 + 12 &= 0 \\ (w-18)^2 &= 400 \\ w-18 &= \pm 20 \\ w &= 38 \quad \text{or} \quad -2 \text{ (rejected)} \end{aligned}$$

OR

$$\begin{aligned} -\frac{3}{100}w^2 + \frac{27}{25}w + \frac{57}{25} &= 0 \\ w^2 - 36w - 76 &= 0 \\ (w-38)(w+2) &= 0 \\ w &= 38 \quad \text{or} \quad w = -2 \text{ (rejected)} \end{aligned}$$

The javelin covered 38 m before hitting the ground.

3 Solve the equation $2 \operatorname{cosec} \theta - 5 \sin \theta = 3 \cot \theta$ for $0^\circ \leq \theta \leq 360^\circ$.

[6]

$2 \operatorname{cosec} \theta - 5 \sin \theta = 3 \cot \theta$ $2 - 5 \sin^2 \theta = 3 \cos \theta$ $2 - 5(1 - \cos^2 \theta) = 3 \cos \theta$ $5 \cos^2 \theta - 3 \cos \theta - 3 = 0$ $\cos \theta = \frac{-(-3) \pm \sqrt{(-3)^2 - 4(5)(-3)}}{2(5)}$ $= \frac{3 \pm \sqrt{69}}{10}$ $= 1.13066 \text{ (rejected)} \quad \text{or} \quad = -0.530662$ $\alpha = 57.9499^\circ$ $\theta = 122.1^\circ, 237.9^\circ$	
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- 4 A curve passes through the point $A(0, -1)$ and the gradient of the curve at this point is parallel to the line $y + 2x = -1$. Given $\frac{d^2y}{dx^2} = 2e^{2x} + 6e^{-2x} + 9\cos 3x$, find y in terms of x . [7]

$$\begin{aligned}\frac{dy}{dx} &= \int (2e^{2x} + 6e^{-2x} + 9\cos 3x) \, dx \\ &= e^{2x} - 3e^{-2x} + 3\sin 3x + c_1\end{aligned}$$

$$\begin{aligned}\text{When } x = 0, \frac{dy}{dx} &= -2, \\ -2 &= 1 - 3 + 0 + c_1 \\ c_1 &= 0\end{aligned}$$

$$\frac{dy}{dx} = e^{2x} - 3e^{-2x} + 3\sin 3x$$

$$\begin{aligned}y &= \int e^{2x} - 3e^{-2x} + 3\sin 3x \, dx \\ &= \frac{1}{2}e^{2x} + \frac{3}{2}e^{-2x} - \cos 3x + c_2\end{aligned}$$

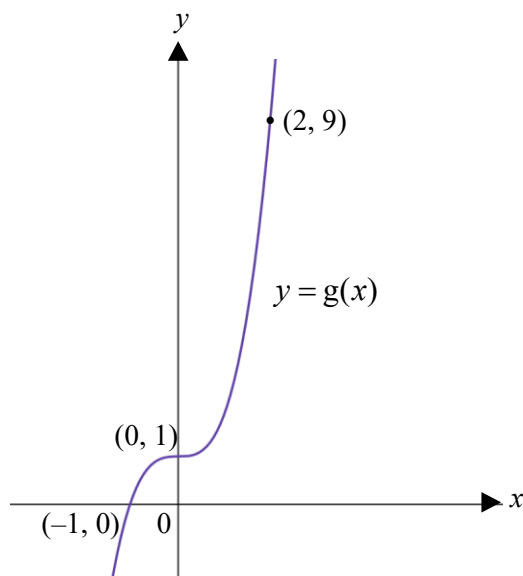
$$\begin{aligned}\text{When } x = 0, y &= -1, \\ -1 &= \frac{1}{2} + \frac{3}{2} - 1 + c_2 \\ c_2 &= -2\end{aligned}$$

$$y = \frac{1}{2}e^{2x} + \frac{3}{2}e^{-2x} - \cos 3x - 2$$

- 5 (a) Given that $\int_3^{10} f(x) \, dx = 15$ and $f(x)$ lies above the x -axis for $x \geq 3$, find the exact value of $\int_3^7 [5x^2 - 3f(x)] \, dx - \int_7^{10} 3f(x) \, dx$. [3]

$ \begin{aligned} & \int_3^7 [5x^2 - 3f(x)] \, dx - \int_7^{10} 3f(x) \, dx \\ &= \int_3^7 5x^2 \, dx - \int_3^7 3f(x) \, dx - \int_7^{10} 3f(x) \, dx \\ &= \int_3^7 5x^2 \, dx - 3 \left[\int_3^7 f(x) \, dx + \int_7^{10} f(x) \, dx \right] \\ &= \int_3^7 5x^2 \, dx - 3 \int_3^{10} f(x) \, dx \\ &= \left[\frac{5}{3} x^3 \right]_3^7 - 3(15) \\ &= \frac{5}{3} (7^3 - 3^3) - 45 \\ &= 481\frac{2}{3} \quad \text{or} \quad \frac{1445}{3} \end{aligned} $	
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- (b) The diagram shows part of the graph of $y = g(x)$.
 Given that the curve passes through the points $(0, 1)$, $(-1, 0)$ and $(2, 9)$, find the value of $\int_0^2 y \, dx + \int_1^9 x \, dy$. [1]



$ \begin{aligned} & \int_0^2 y \, dx + \int_1^9 x \, dy \\ &= 2 \times 9 \\ &= 18 \end{aligned} $	
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- 6 (a) Find the value of a for which $2x-3$ is a factor of $P(x) = 2x^3 - 5x^2 + ax + 6$, where a is an integer. Hence, determine whether $x+2$ is a factor of $P(x)$. [4]

$P\left(\frac{3}{2}\right) = 0$ $2\left(\frac{3}{2}\right)^3 - 5\left(\frac{3}{2}\right)^2 + a\left(\frac{3}{2}\right) + 6 = 0$ $\frac{27}{4} - \frac{45}{4} + \frac{3}{2}a + 6 = 0$ $\frac{3}{2}a = -\frac{3}{2}$ $a = -1$ $\therefore P(x) = 2x^3 - 5x^2 - x + 6$ Since $P(-2) = 2(-2)^3 - 5(-2)^2 - (-2) + 6$ $= -28 \neq 0$ Hence, $x+2$ is not a factor of $P(x)$.	
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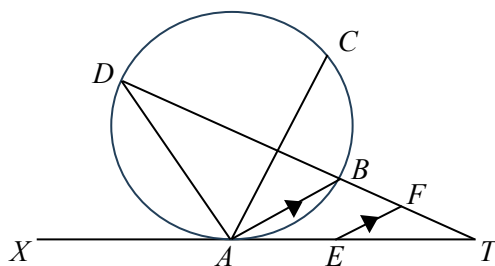
- (b) Factorise $P(x)$ completely. [2]

By observation, $P(x) = 2x^3 - 5x^2 - x + 6$ $= (2x-3)(x^2 + bx - 2)$ Comparing x terms, $-3b - 4 = -1$ $b = -1$	OR By Long division, $\begin{array}{r} x^2 - x - 2 \\ 2x-3 \overline{) 2x^3 - 5x^2 - x + 6} \\ \underline{-(2x^3 - 3x^2)} \\ -x^2 - x + 6 \\ \underline{-(-2x^2 + 3x)} \\ -4x + 6 \\ \underline{-(-4x + 6)} \\ 0 \end{array}$	
$P(x) = (2x-3)(x^2 - x - 2)$ $= (2x-3)(x-2)(x+1)$		

- (c) Using your answers in (b), solve the equation $2^{3y+1} - 5(2^{2y}) = 2^y - 6$. [3]

$2^{3y+1} - 5(2^{2y}) = 2^y - 6$ $2(2^y)^3 - 5(2^y)^2 - 2^y + 6 = 0$ $(2^y - 3)(2^y - 2)(2^y + 1) = 0$ $\therefore 2^y = \frac{3}{2} \quad \text{or} \quad 2^y = 2 \quad \text{or} \quad 2^y = -1 \text{ (rejected, } 2^y > 0)$ $y = \frac{\ln \frac{3}{2}}{\ln 2} \quad y = 1$ $= 0.585 \text{ (to 3s.f.)}$	
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7



In the diagram, A, B, C and D lie on a circle. XT is the tangent to the circle at A and the line DB extended meets the tangent at T . Points E and F lie on XAT and DBT respectively. EF is parallel to AB and point E is the mid-point of AT . It is given that AB bisects angle CAT and AD bisects angle CAX .

(a) Prove that $AB = BC$.

[2]

Let $\angle BAT = \angle BAC = b$ (AB bisects $\angle CAT$)
 $\angle BAT = \angle ACB = b$ ($\angle$ s in alt. segments)
 $\therefore \angle BAC = \angle ACB = b$
 $\triangle ABC$ is isosceles with base angles $\angle BAC$ and $\angle ACB$,
 hence $AB = BC$.

OR

Let $\angle BAT = \angle BAC = b$ (AB bisects $\angle CAT$)
 $\angle ADB = \angle BAT = b$ ($\angle$ s in alt. segments)
 $\angle ADB = \angle ACB = b$ ($\angle$ s in the same segment)
 $\therefore \angle BAC = \angle ACB = b$
 $\triangle ABC$ is isosceles with base angles $\angle BAC$ and $\angle ACB$,
 hence $AB = BC$.

(b) Prove that BD is the diameter of the circle.

[3]

$\angle CAX = 180^\circ - 2b$ (adj. $\angle$ s on a str. line)
 Since AD bisects $\angle CAX$,
 $\angle CAD = \frac{180^\circ - 2b}{2}$
 $= 90^\circ - b$
 $\angle BAD = \angle CAD + \angle BAC$
 $= 90^\circ - b + b$
 $= 90^\circ$
 By property of right angle in a semicircle, since $\angle BAD = 90^\circ$,
 BD is the diameter of the circle.

OR

Let $\angle DAX = \angle DAC = a$ (AD bisects $\angle CAX$) $\angle CAX + \angle CAT = 180^\circ$ (adj. $\angle$s on a str. line) $2a + 2b = 180^\circ$ $a + b = 90^\circ$ $\angle DAB = \angle DAC + \angle BAC$ $= a + b$ $= 90^\circ$ $\therefore$ By property of right angle in a semi-circle, since $\angle DAB = 90^\circ$, BD is a diameter of the circle.	
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- (c) (i) Prove that triangle TAD is similar to triangle TBA . [3]

$\angle ATD = \angle BTA$ (common angle) $\angle TDA = \angle TAB$ ($\angle$s in alt. segment) By AA Similarity Test, $\triangle TAD$ is similar to $\triangle TBA$. OR $\angle ADT = \angle BAT = b$ ($\angle$s in alt. segments) $\angle ATD = \angle BTA$ (common angle) $= 180^\circ - a - 3b$ (angle sum of $\triangle TAD$) $\angle TAD = a + 2b$ $\angle TBA = 180^\circ - b - (180^\circ - a - 3b)$ (angle sum of a $\triangle TBA$) $= a + 2b$ $\therefore \angle TAD = \angle TBA$ By AA Similarity Test, $\triangle TAD$ is similar to $\triangle TBA$.	
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- (ii) Hence, show that $BT \times DT = 4ET^2$. [2]

Since $\triangle TAD$ is similar to $\triangle TBA$, then $\frac{AT}{BT} = \frac{DT}{AT}$ Since $AT = 2ET$ (given), $\frac{2ET}{BT} = \frac{DT}{2ET}$ $BT \times DT = 4ET^2 \text{ (shown)}$	
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8 Given $y = e^{-2x}x^2$,

(a) y is a solution of the equation $e^{2x} \left(4y + 4 \frac{dy}{dx} + \frac{d^2y}{dx^2} \right) = k$, where k is a constant.

Show that $k = 2$.

[6]

$$\begin{aligned} \frac{dy}{dx} &= x^2(-2e^{-2x}) + e^{-2x}(2x) \\ &= 2e^{-2x}(x - x^2) \end{aligned}$$

$$\begin{aligned} \frac{d^2y}{dx^2} &= 2e^{-2x}(1 - 2x) + (x - x^2)(-4e^{-2x}) \\ &= 2e^{-2x} - 8e^{-2x}x + 4e^{-2x}x^2 \end{aligned}$$

$$\begin{aligned} k &= e^{2x} \left(4y + 4 \frac{dy}{dx} + \frac{d^2y}{dx^2} \right) \\ &= e^{2x} \left\{ 4(e^{-2x}x^2) + 4[2e^{-2x}(x - x^2)] + 2e^{-2x} - 8e^{-2x}x + 4e^{-2x}x^2 \right\} \\ &= 2 \end{aligned}$$

- (b) find the range of values of x for which y increases as x increases.

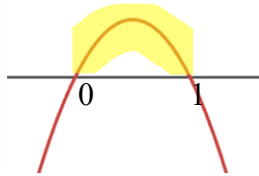
[3]

For increasing function, $\frac{dy}{dx} > 0$

Since $2e^{-2x} > 0$, then $x - x^2 > 0$

$$x(1-x) > 0$$

$$0 < x < 1$$



9 It is given that $\lg(2a+b) = \lg a - \lg b$.

(a) (i) Express a in terms of b .

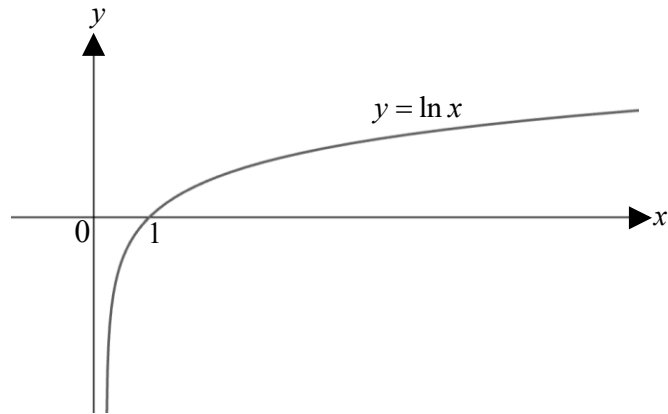
[3]

$\lg(2a+b) = \lg a - \lg b$ $\lg(2a+b) = \lg \frac{a}{b}$ $2a+b = \frac{a}{b}$ $2ab+b^2 = a$ $b^2 = a - 2ab$ $= a(1-2b)$ $a = \frac{b^2}{1-2b}$	
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(ii) State the range of values of a and explain clearly why $0 < b < \frac{1}{2}$.

[2]

(1) For $\lg a$ to be defined, $a > 0$ $\therefore \frac{b^2}{1-2b} > 0$ Since $b^2 \geq 0$ for all real values of b, then $1-2b > 0$ and $b < \frac{1}{2}$ (2) For $\lg b$ to be defined, $b > 0$ $\therefore 0 < b < \frac{1}{2}$	
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- (b) The diagram shows part of the curve $y = \ln x$ for $x > 0$.
Use the graph to explain how you would obtain a graphical solution for the equation $x^2 = e^{4-x}$. [3]

$$x^2 = e^{4-x}$$

$$x = \left(e^{4-x} \right)^{\frac{1}{2}}$$

$$x = e^{2-\frac{1}{2}x}$$

$$\ln x = \ln e^{2-\frac{1}{2}x}$$

$$y = 2 - \frac{1}{2}x$$

Draw the line $y = 2 - \frac{1}{2}x$ on the same axes.

The **x-coordinate** of the point of intersection between the two graphs is the solution to the equation.

- 10** A particle is travelling in a straight line. Its velocity, v m/s, where t is the time in seconds after passing a fixed point O , is given by $v = 3 - 6\sin 3t$.

(a) Determine the range of values of v . [1]

$-3 \leq v \leq 9$	
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(b) Find the exact values of t for the first two times the particle is at instantaneous rest. [3]

When $v = 0$, $\sin 3t = \frac{1}{2}$ $3t = \frac{\pi}{6}, \frac{5\pi}{6}$ $t = \frac{\pi}{18}, \frac{5\pi}{18}$ The particle will be at instantaneous rest at $t = \frac{\pi}{18}$ and $\frac{5\pi}{18}$.	
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- (c) Find the displacement of the particle when $t = \frac{5\pi}{18}$.

[3]

$$s = \int 3 - 6 \sin 3t \, dt$$

$$= 3t + 2 \cos 3t + c$$

When $t = 0, s = 0 \therefore c = -2$

$$\therefore s = 3t + 2 \cos 3t - 2$$

When $t = \frac{5\pi}{18}, s = -1.11$

The displacement is -1.11 m.

- (d) Find the distance travelled by the particle for the first $\frac{\pi}{2}$ seconds.

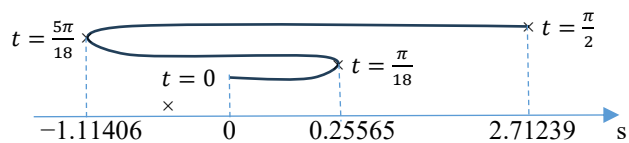
[2]

When $t = 0, s = 0$

When $t = \frac{\pi}{18}, s = 0.25565$

When $t = \frac{5\pi}{18}, s = -1.11406$

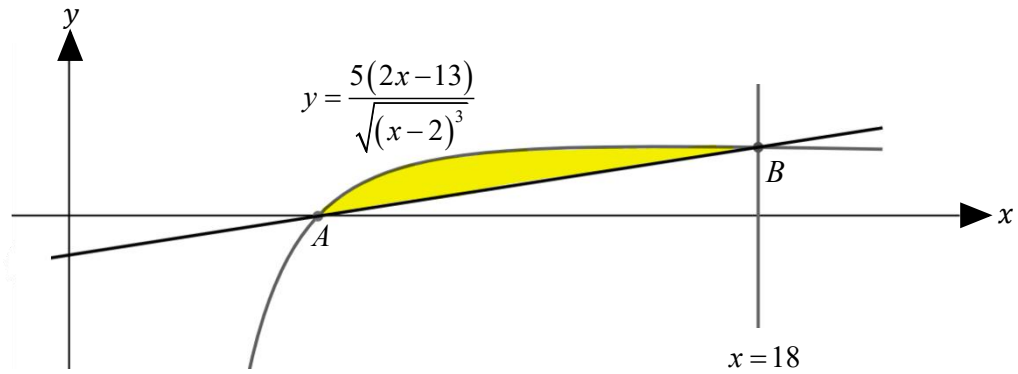
When $t = \frac{\pi}{2}, s = 2.71239$



$$\begin{aligned} \text{Total distance} &= 2(0.25565) + 2(1.11406) + 2.71239 \\ &= 5.45181 \\ &= 5.45 \text{ m} \end{aligned}$$

- 11 (a) Show that $\frac{d}{dx} \left(\frac{2x+5}{\sqrt{x-2}} \right) = \frac{2x-13}{2\sqrt{(x-2)^3}}$. [3]

$\begin{aligned} & \frac{d}{dx} \left(\frac{2x+5}{\sqrt{x-2}} \right) \\ &= \frac{(x-2)^{\frac{1}{2}}(2) - (2x+5)\left(\frac{1}{2}\right)(x-2)^{-\frac{1}{2}}}{x-2} \\ &= \frac{\frac{1}{2}(x-2)^{-\frac{1}{2}}[4(x-2) - (2x+5)]}{x-2} \\ &= \frac{4x-8-2x-5}{2\sqrt{(x-2)^3}} \\ &= \frac{2x-13}{2\sqrt{(x-2)^3}} \end{aligned}$	
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- (b) The diagram shows part of the curve $y = \frac{5(2x-13)}{\sqrt{(x-2)^3}}$ where $x > 0$. The curve intersects the x-axis at point A. The line $x = 18$ cuts the curve at point B. Find, using part (a), the area bounded by the curve and line AB. Show all necessary working. Leave your answer correct to 2 decimal places. [6]

$$\text{When } y = 0, x = 6\frac{1}{2}$$

$$\therefore A\left(6\frac{1}{2}, 0\right)$$

$$\text{When } x = 18, y = 1\frac{51}{64} \text{ or } \frac{115}{64}$$

$$\therefore B\left(18, 1\frac{51}{64}\right)$$

$$\begin{aligned} \text{Area of triangle} &= \frac{1}{2} \left(1\frac{51}{64}\right) \left(18 - 6\frac{1}{2}\right) \\ &= \frac{2645}{256} \text{ units}^2 \text{ or } 10.3320 \text{ units}^2 \end{aligned}$$

$$\int_{6\frac{1}{2}}^{18} \frac{5(2x-13)}{\sqrt{(x-2)^3}} dx$$

$$= \int_{6\frac{1}{2}}^{18} 10 \left(\frac{(2x-13)}{2\sqrt{(x-2)^3}} \right) dx$$

$$= 10 \left[\frac{(2x+5)}{\sqrt{(x-2)}} \right]_{6\frac{1}{2}}^{18}$$

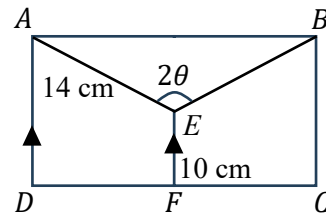
$$= 17.6471$$

$$\text{Required area} = 17.6471 - \frac{2645}{256}$$

$$= 7.3151$$

$$= 7.32 \text{ units}^2$$

12



In the diagram, $ABCFD$ shows the design of the front of a handbag. $ABCFD$ is a rectangle, triangle ABE is isosceles with $AE = BE = 14$ cm and $\angle AEB = 2\theta$ radians. EF is parallel to AD and $EF = 10$ cm. The perimeter of the handbag is P cm.

- (a) Show that P can be expressed as $20 + 28\cos\theta + 56\sin\theta$. [2]

Let X be the foot of the perpendicular from E to AB .

$$AX = 14\sin\theta$$

$$EX = 14\cos\theta$$

$$\begin{aligned} P &= 2(10) + 2(14\cos\theta) + 4(14\sin\theta) \\ &= 20 + 28\cos\theta + 56\sin\theta \end{aligned}$$

- (b) Express P in the form $20 + R\cos(\theta - \alpha)$, where $R > 0$ and $0 < \alpha < \frac{\pi}{2}$. [3]

$$\begin{aligned} R &= \sqrt{(56)^2 + (28)^2} \\ &= \sqrt{3920} \text{ or } 28\sqrt{5} \end{aligned}$$

$$\begin{aligned} \alpha &= \tan^{-1} \frac{56}{28} \\ &= 1.1071 \text{ rad} \end{aligned}$$

$$\therefore P = 20 + 28\sqrt{5}\cos(\theta - 1.11) \text{ (to 3 s.f.)}$$

- (c) The perimeter of the handbag is lined with a chain of pearls. Given that θ can vary and that the designer paid \$110 for the pearls, find the minimum cost per metre of chained pearls and the corresponding value of θ . [3]

$\begin{aligned}\text{Minimum cost per metre} &= \frac{110}{\text{Maximum } P} \\ &= \frac{110}{(20 + 28\sqrt{5}) \div 100} \\ &= \$133.16\end{aligned}$ when $\cos(\theta - 1.1071) = 1$ $\theta - 1.1071 = 0$ $\theta = 1.11$	
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- (d) Explain whether P can be equal to 20 cm. [2]

When $P = 20$, $\cos(\theta - 1.1071) = 0$ $\theta - 1.1071 = \frac{\pi}{2}$ $\theta = 2.67 > \frac{\pi}{2}$ Since it is not possible for $\theta > \frac{\pi}{2}$, hence it is not possible for $P = 20$.	
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